

Week 2 — Data types & central tendency

Business Forecasting · LIVE pilot (slides + exercises in one deck)

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Where we are

C1 Framing → C2 Describing data → C3 Uncertainty → C4–C5 Regression → C6 Forecasting

We are in **C2 — describing data**. This week has two halves: first **what kind of data** we have (types & structure), then the **summary numbers** that describe its center — mean, median, mode, and percentiles.

This deck is the *pilot* of the integrated format: the exercises you used to open in a separate learnr window now appear **as slides**, and the code runs right here in the browser.

Part 1 — What structure is the data?

Unit of observation

Every dataset has a **unit of observation** — *what does one row represent?* Get this wrong and every number after it is wrong.

Ask: *a worker? a firm? a worker-month? a transaction?*

The structure of the data follows from **how the unit is observed over time**. Three cases — let's see each in a real table.

Cross-sectional data

Many units, **one point in time**. “Who is where, right now?”

Customer ID	Satisfaction (1-10)	Repurchase?
1	7	Likely
2	8	Unlikely
3	5	Likely
4	9	Likely

Example: a one-time customer survey. *Business use:* “What drives satisfaction **now**?”

Time-series (longitudinal) data

One unit, many time points. *“Is it trending?”*

Year	Revenue (MXN)	Employees
2018	50000	50
2019	52000	55
2020	55000	60
2021	58000	65
2022	60000	70

Example: one company’s revenue across years. *Business use:* “Forecast **next year’s** revenue.”

Panel data

Many units, followed over time — cross-section × time-series.

Year	Company	Revenue (MXN)
2020	A	50000
2020	B	60000
2020	C	70000
2021	A	52000
2021	B	63000
2021	C	72000
2022	A	55000
2022	B	65000
2022	C	75000

Example: several companies, each across years. *Business use:* “How did **each** firm grow?”

Your turn — name the structure

Month	Coin	Market cap
Jan	Bitcoin	60000
Jan	Ethereum	40000
Jan	Dogecoin	10000
Feb	Bitcoin	62000
Feb	Ethereum	41000
Feb	Dogecoin	10500

Three coins, tracked across months — which structure?

Panel — multiple subjects (coins) × multiple time points.

Part 2 — What kind of variables?

Meet our running dataset: the workforce

We'll use one business dataset all week — a sample of the Mexican labor market, **3,000 workers in 2022**. You're a **compensation analyst**: *is pay fair, and how does it vary?*

Monthly income (MXN)	Age	Hours/week	Gender	Region	Age group
10000	26	20	Men	Rest of Mexico	18-29
30100	58	70	Men	Rest of Mexico	50-65
6000	30	49	Men	Rest of Mexico	30-39
3870	43	9	Women	Rest of Mexico	40-49
7740	48	50	Women	Rest of Mexico	40-49
5160	35	51	Men	Rest of Mexico	30-39

First job: know what *kind* of variable each column is.

Categorical variables

Labels / groups. Summarize with **counts and shares**.

Nominal — no natural order

- **gender**: Women, Men
- **region**: Mexico City, Rest of Mexico
- Social platform: Facebook, TikTok, ...

Ordinal — ordered, but distances aren't meaningful

- Education: primary < high school < university
- Satisfaction: Very unsatisfied < ... < Very satisfied

We can *rank* ordinal categories, but “Satisfied – Neutral” is not a real quantity.

Numerical variables

Quantities you can do arithmetic on. Summarize with **averages and spread**.

Discrete — only certain values

- `age` (whole years)
- Number of vehicles a household owns
- App downloads

Continuous — any value in a range

- `monthly_income` (MXN)
- `hours_worked`
- Exchange rate MXN/USD

Ordinal vs discrete: both use numbers, but ordinal *labels* have no arithmetic meaning — discrete *counts* do.

Discrete = a countable set of values

A discrete variable can only land on **specific values** — nothing in between.

- Number of children in a household
- App downloads from the store
- Coke product sizes: 0.33L, 0.5L, 1L, 2.25L



So — what is each of our variables?

monthly_income · age · hours_worked · gender · region

Numerical or categorical? Discrete or continuous, nominal or ordinal?

Don't answer from the names. **Get the evidence out of R first.**



Exercise 1 — Look at the data

Show the first 10 rows of `workforce`, then its structure.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 # Your code here
```



Exercise 2 — Classify all five

Now classify each of the five variables. `str()` gives you a first answer — confirm it by asking **how many distinct values** each variable takes, and what they look like.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 # How many distinct values does each variable take?
```

```
2
```

```
3 # Look at the actual values of the ones with only a few:
```

Classify our workforce variables

Variable	Type	Subtype
monthly_income	Numerical	Continuous
age	Numerical	Discrete
hours_worked	Numerical	Discrete
gender	Categorical	Nominal
region	Categorical	Nominal

The type decides the summary and the chart — get it wrong and your client gets a misleading number.

Part 3 — Visualizing categorical data

How is the workforce split across age groups?

The first summary of any categorical variable is a **frequency table**: the count n_i in each category, and the relative frequency $p_i = n_i / N$.

Build it before you see it.



Exercise 3 — Build the frequency table

Make a frequency table of `age_group` — counts first, then shares. Then do the same for `region`.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 # Your code here
```

Frequency tables — the polished version

Age group	Workers	Share
18-29	654	21.8%
30-39	883	29.4%
40-49	815	27.2%
50-65	648	21.6%

Region	Workers	Share
Mexico City	59	2.0%
Rest of Mexico	2941	98.0%

Categories are **mutually exclusive** and **collectively exhaustive**, so the shares always sum to 1. Exact — but a manager grasps a **picture** faster.

The table is exact — the picture is faster

A manager will not read a frequency table. Which picture lets them compare age groups at a glance?

Before we show you one: **build it yourself.**



Exercise 4 — Build the chart

Draw the number of workers in each `age_group`. Then do the same for `region` — does the same recipe work for a variable with no natural order?

Exercise

↺ Start Over

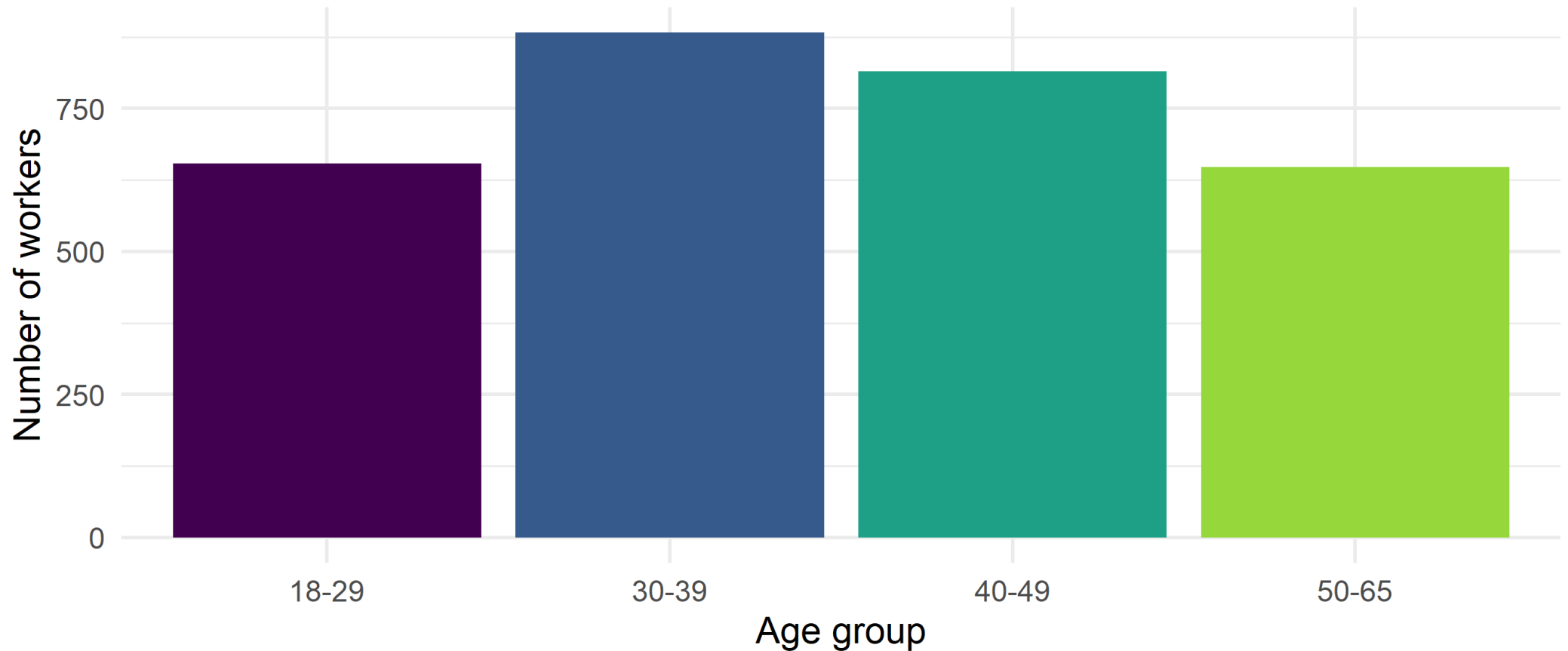
💡 Show Hint

▶ Run Code

```
1 # barplot() takes a TABLE of counts, not the raw column.
```

Bar chart — the polished version

Most of the sampled workforce is between 30 and 49



Bars are the right tool for categories: length = count, easy to compare.

Now compare women and men

Are women and men spread across age groups differently? Plot the **counts** by gender and you will conclude “there are more men everywhere” — which is true of the sample, and tells you nothing about the age profile.

What should you plot instead?



Exercise 5 — Make the comparison fair

Build the chart that compares the age profile of women and men **fairly**. Start from the two-way table, then fix the units.

Exercise

↺ Start Over

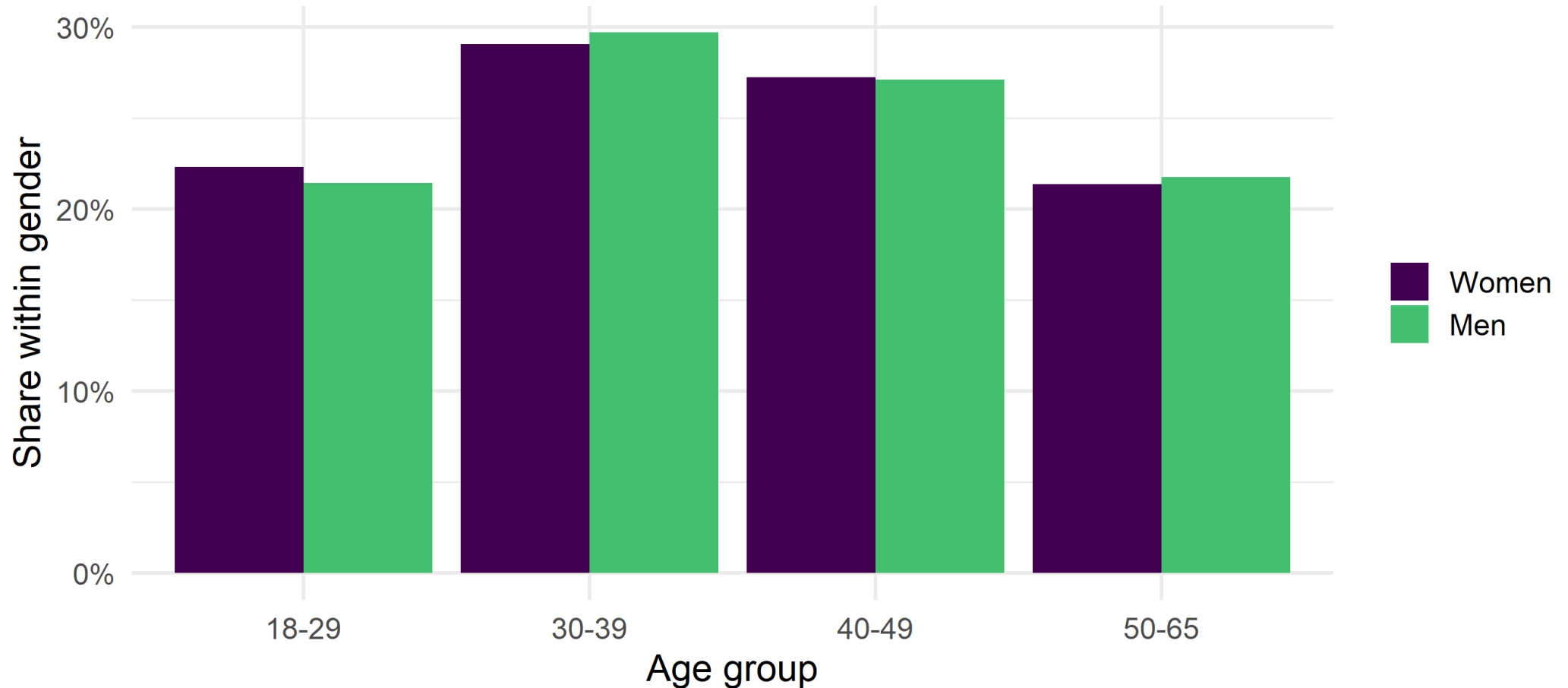
💡 Show Hint

▶ Run Code

```
1 tab <- table(workforce$gender, workforce$age_group)
2 tab
3
4 # Counts are unfair here. Turn each ROW into shares, then plot.
```

Relative frequencies: compare groups fairly

Age profile is similar for women and men



The other option: a pie chart

Same information, drawn as slices instead of bars. Build both and judge for yourself which one you can actually read.



Exercise 6 — Pie vs bars, decide

Draw `age_group` as a **pie** and as a **bar chart**. Then answer: in which one can you rank the four groups faster, and which two groups are hardest to tell apart in the pie?

Exercise

↺ Start Over

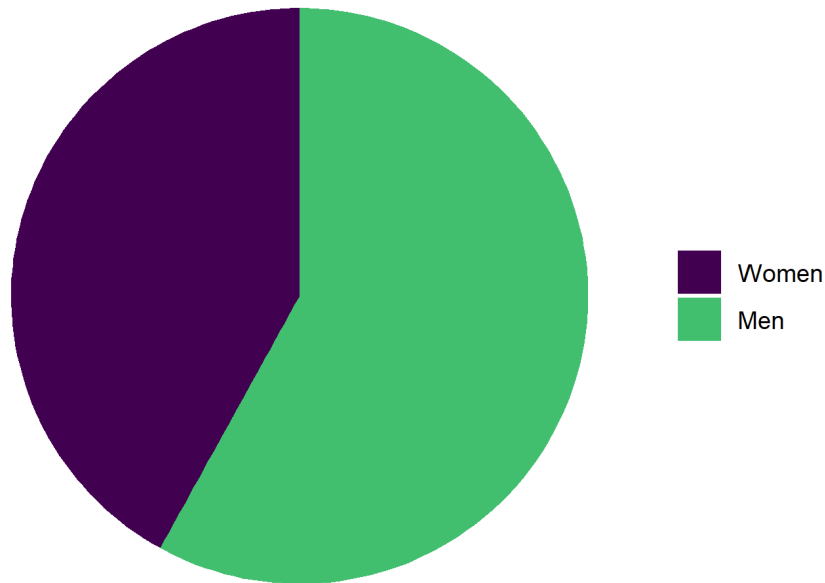
💡 Show Hint

▶ Run Code

```
1 # Your code here
```

Pie charts: usually the wrong choice

Share by gender



People compare **lengths** well and **angles** badly.

- OK for 2–3 slices summing to 100% (like this one).
- For more, a **bar chart** wins.

Reaching for a pie with 5+ slices? Use bars.

Part 4 — Summary statistics: central tendency

Now boil the distribution down to a few numbers managers ask for.

Mean

- **Mean** is the arithmetic average — intuitively, the *balancing point* of the distribution.
- Population mean (N = population size):

$$\mu = E(X) = \frac{\sum_{i=1}^N x_i}{N}$$

- Sample mean (n = sample size) is the sample equivalent:

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n} = \frac{x_1 + x_2 + \dots + x_{n-1} + x_n}{n}$$



Exercise 7 — The mean

(a) Compute the mean `monthly_income` with `mean()`, then verify it by computing `sum(...) / length(...)` yourself — same number.

Exercise [↺ Start Over](#) [⚠ Show Solution](#)

[▶ Run Code](#)

```
1 # Your code here
```

(b) Compute the mean of the binary `gender == "Women"` — what does that number represent?

Exercise [↺ Start Over](#) [⚠ Show Solution](#)

[▶ Run Code](#)

```
1 # Your code here
```

Median

- **Median** represents the middle value when data is sorted
- Half of observations are below it, half are above it.
- For odd size n : the $\frac{n+1}{2}$ -th value. For even n : average of the $\frac{n}{2}$ -th and $\frac{n}{2} + 1$ -th values.

Day	Customers
1	20
2	18
3	25
4	22
5	30
6	21
7	27

$n = 7$ (odd), so:

- Sort: 18, 20, 21, 22, 25, 27, 30.
- Median = the $\frac{7+1}{2} = 4$ -th value.
- **Median = 22.**



Exercise 8 — Mean vs median

Compute both the mean and the median of `monthly_income`. Income is right-skewed — which is larger, and why?

Exercise

↺ Start Over

⚠ Show Solution

▶ Run Code

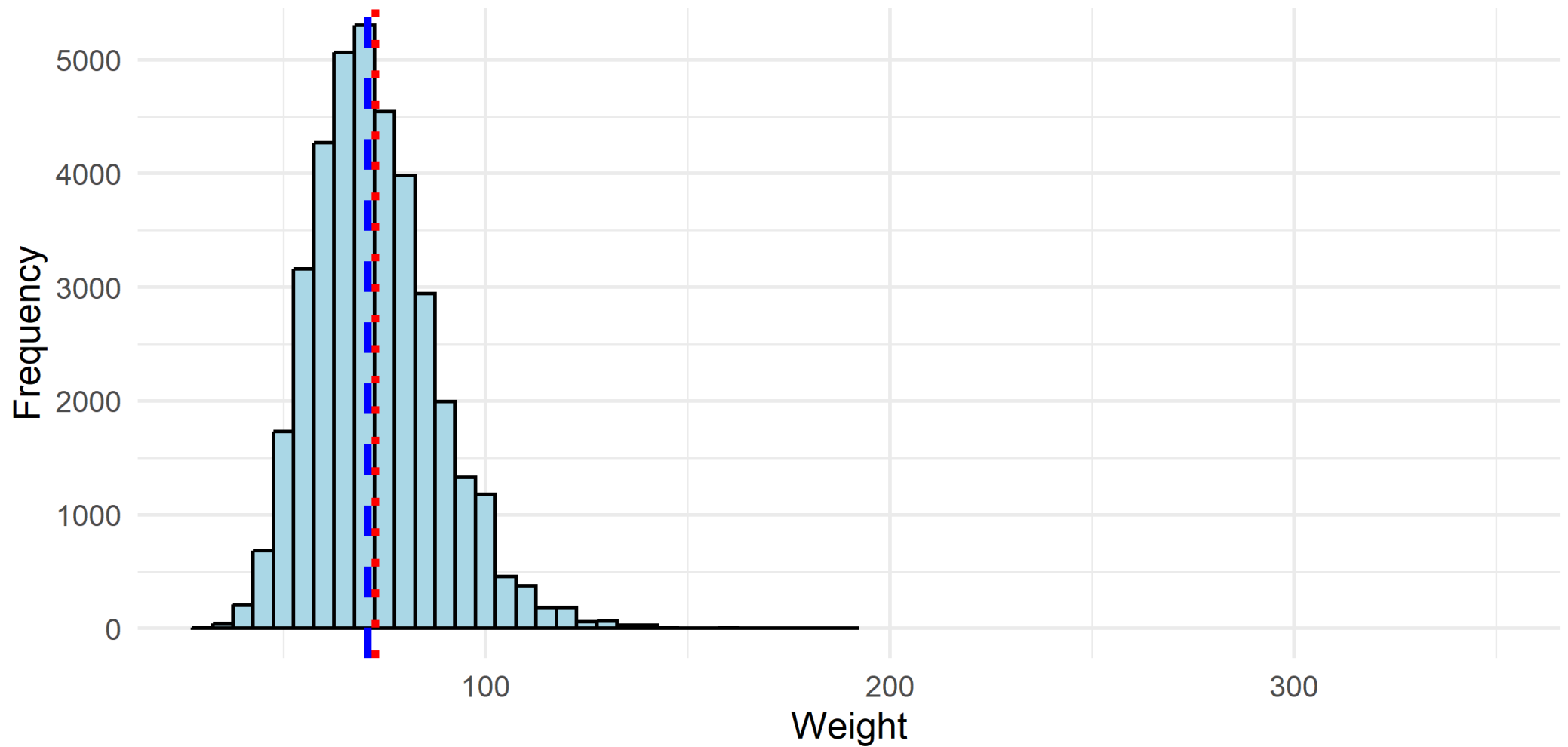
```
1 # Your code here
```

Mean vs Median



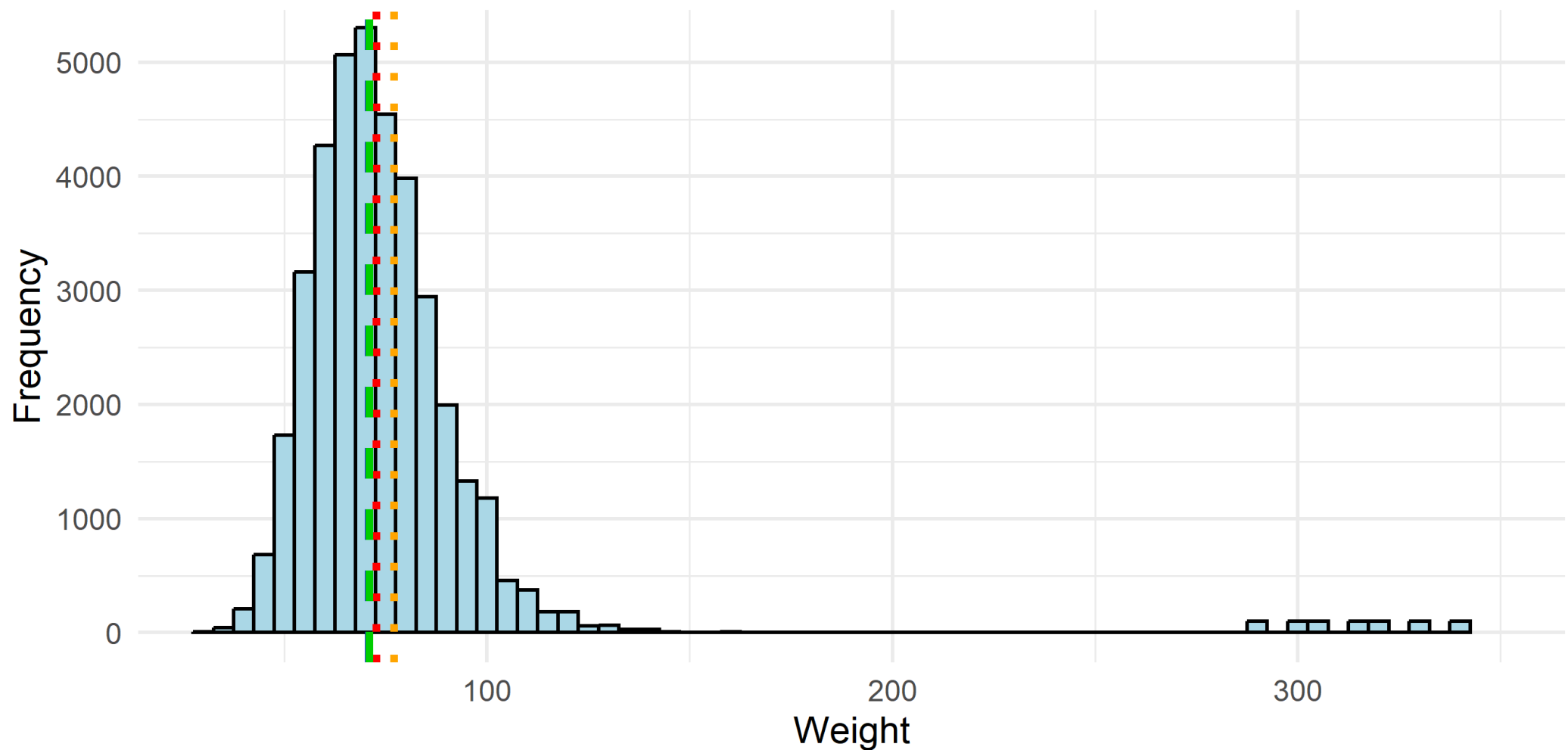
Mean vs median: what an outlier does

Weight in our health data — **mean** (dotted) and **median** (dashed):



- Mean: 72.66, Median: 70.75

Now add outliers on the right tail



- Old Mean: 72.66, **New Mean: 77.05** — dragged up by the outliers
- Old Median: 70.75, **New Median: 70.95** — barely moves

Side note on the Mode

Mode is the most frequent value in the data — the only centre measure that also works for labels, where “average” and “middle” mean nothing.



Exercise 9 — Find the mode

Which `age_group` is the most common? And which `region`? Get R to tell you the *name* of the winner, not just the counts.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

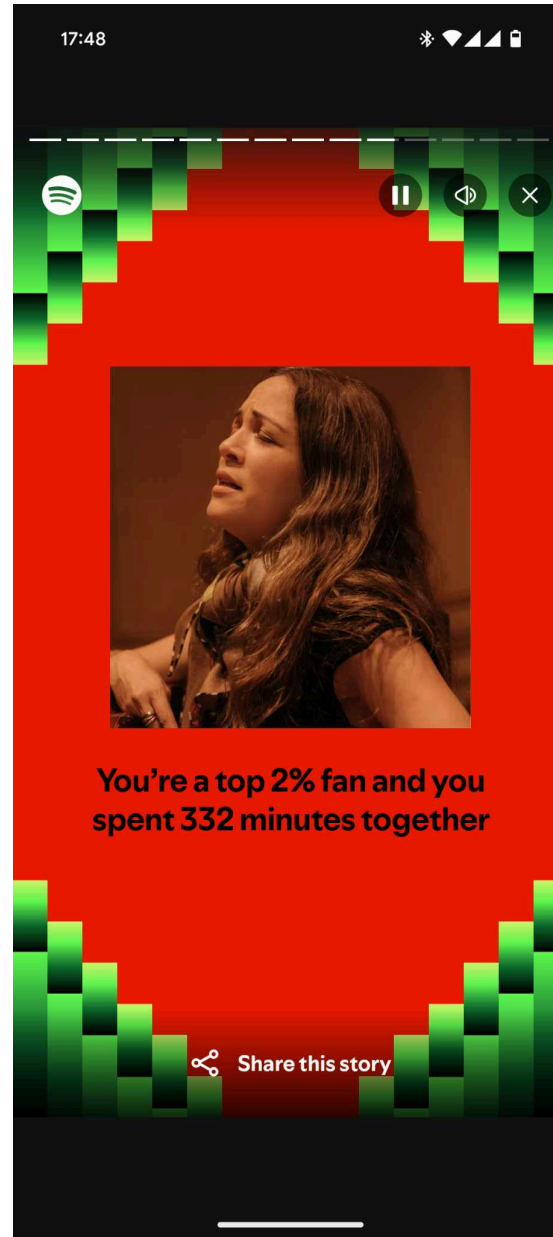
```
1 # Your code here
```

The mode in a frequency table

Age of people with diabetes — the modal age is the row with the largest n_i :

Age	n_i	p_i
60	133	0.032
65	129	0.031
56	124	0.030
58	122	0.029
59	119	0.029
68	116	0.028

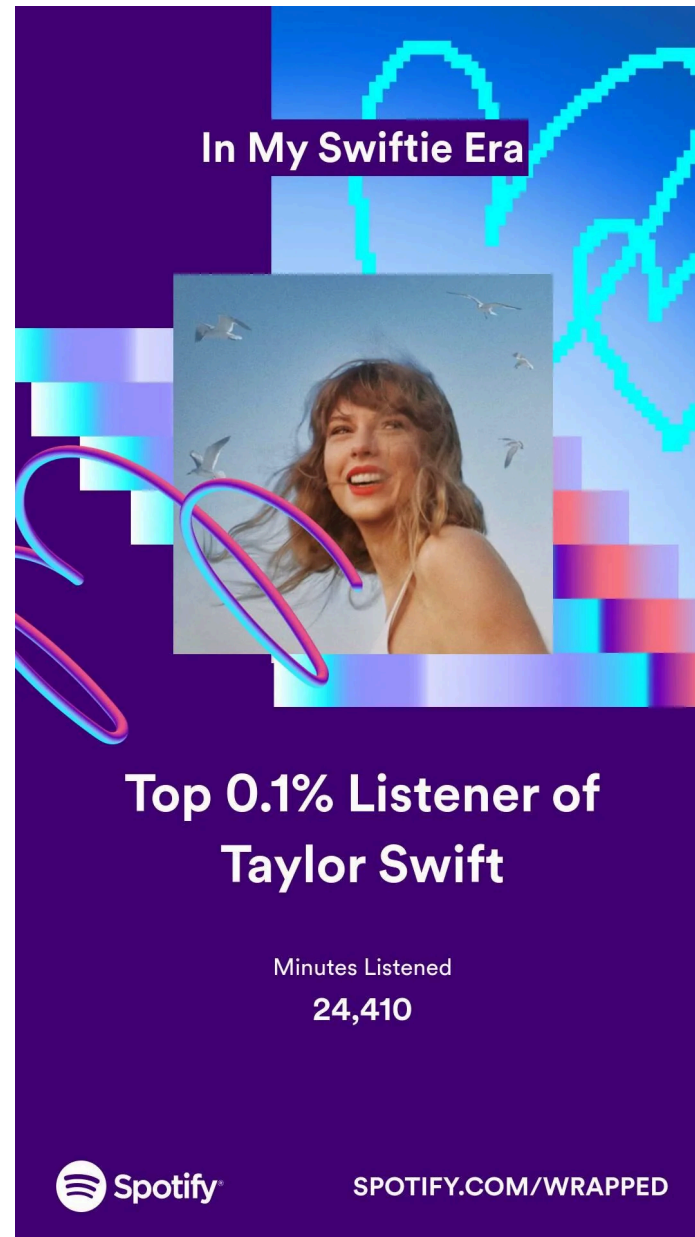
Percentiles



Percentiles



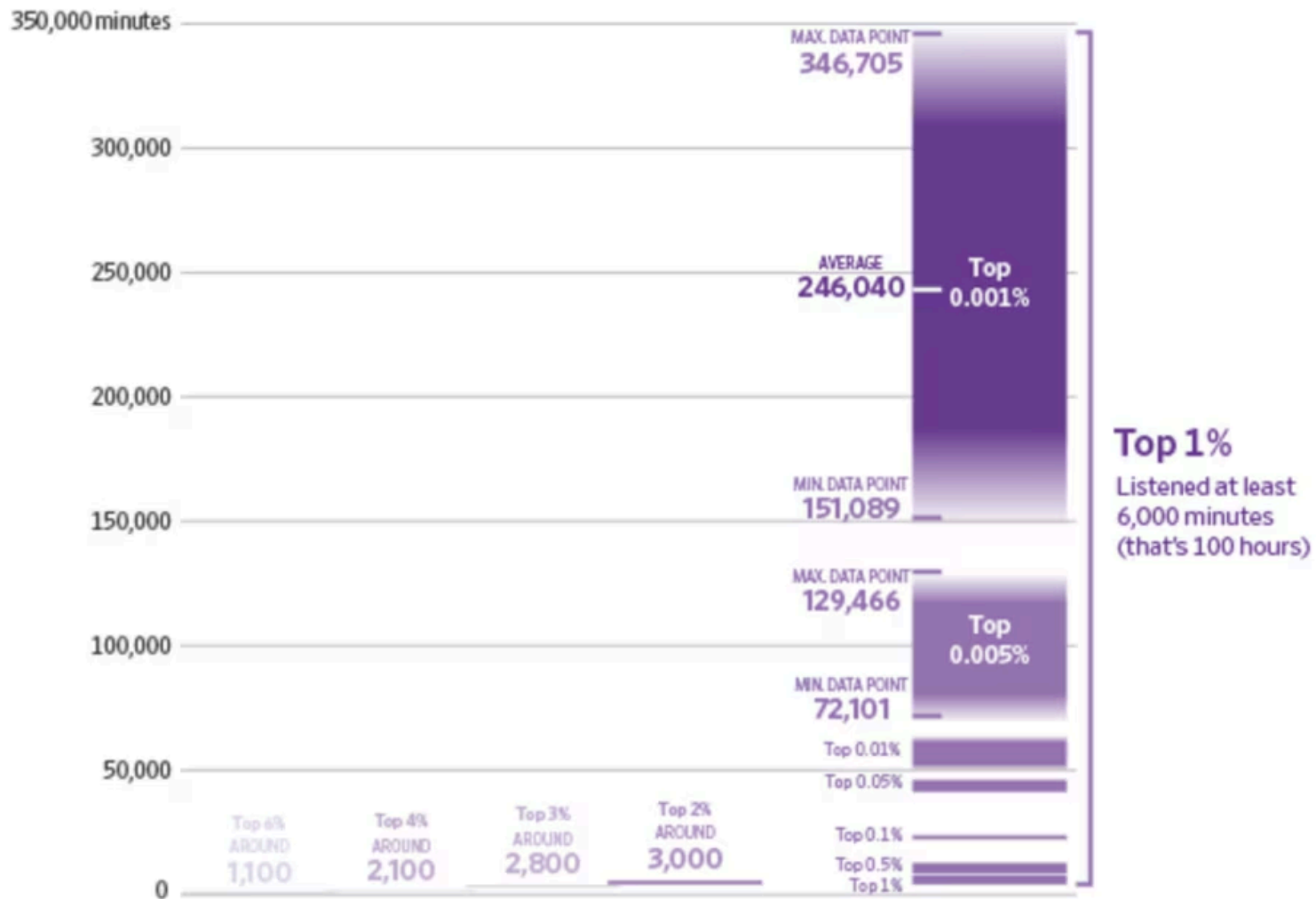
Percentiles



Percentiles



Percentiles



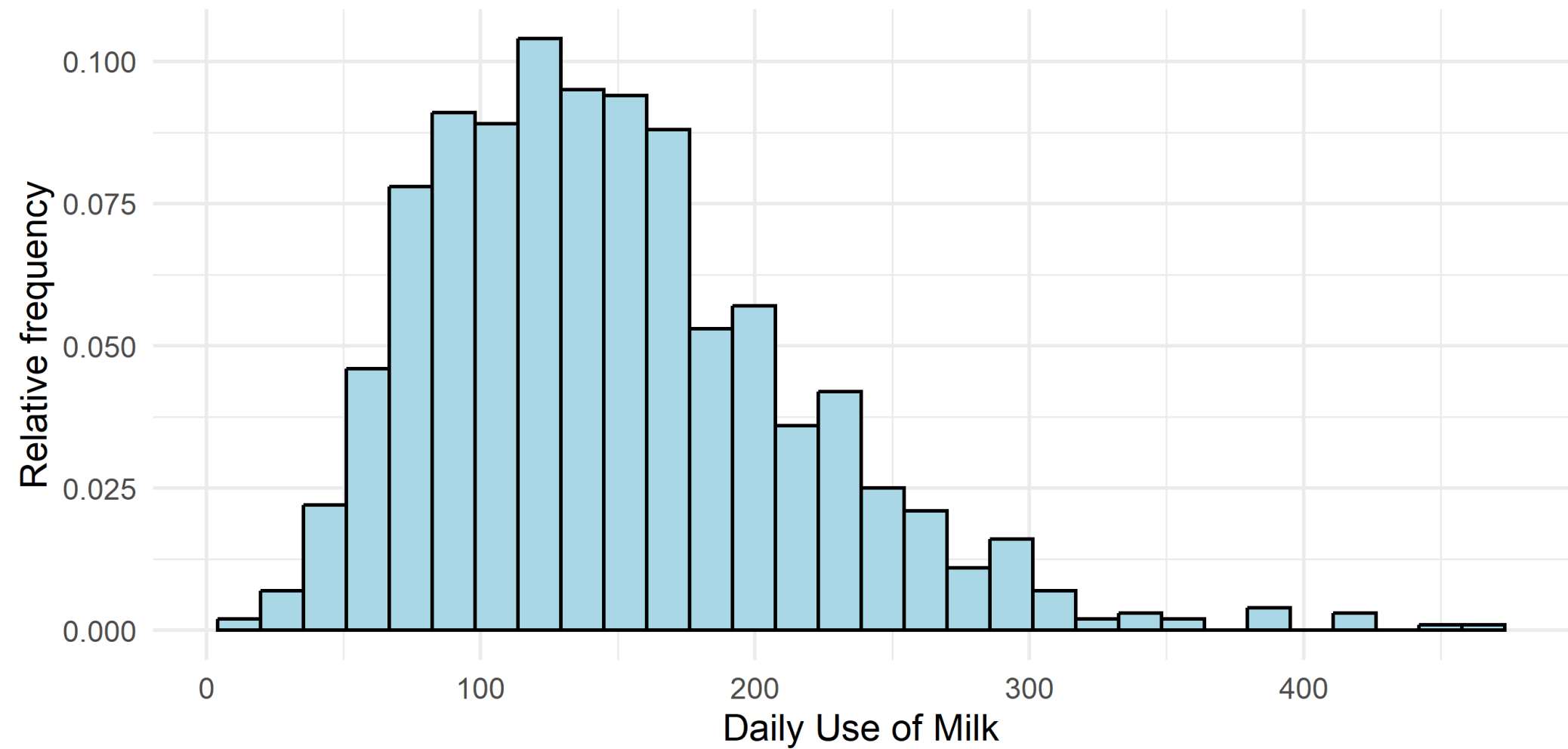
Credits: Wall Street Journal — [Article Link](#)

Percentiles

- How much inventory of milk you need to keep in your Starbucks?
- What is the tradeoff of keeping too much vs too little inventory?
- Suppose we want to have enough milk to cover sales on 95% of days
- To figure it out, let's look at the distribution of the daily use of milk

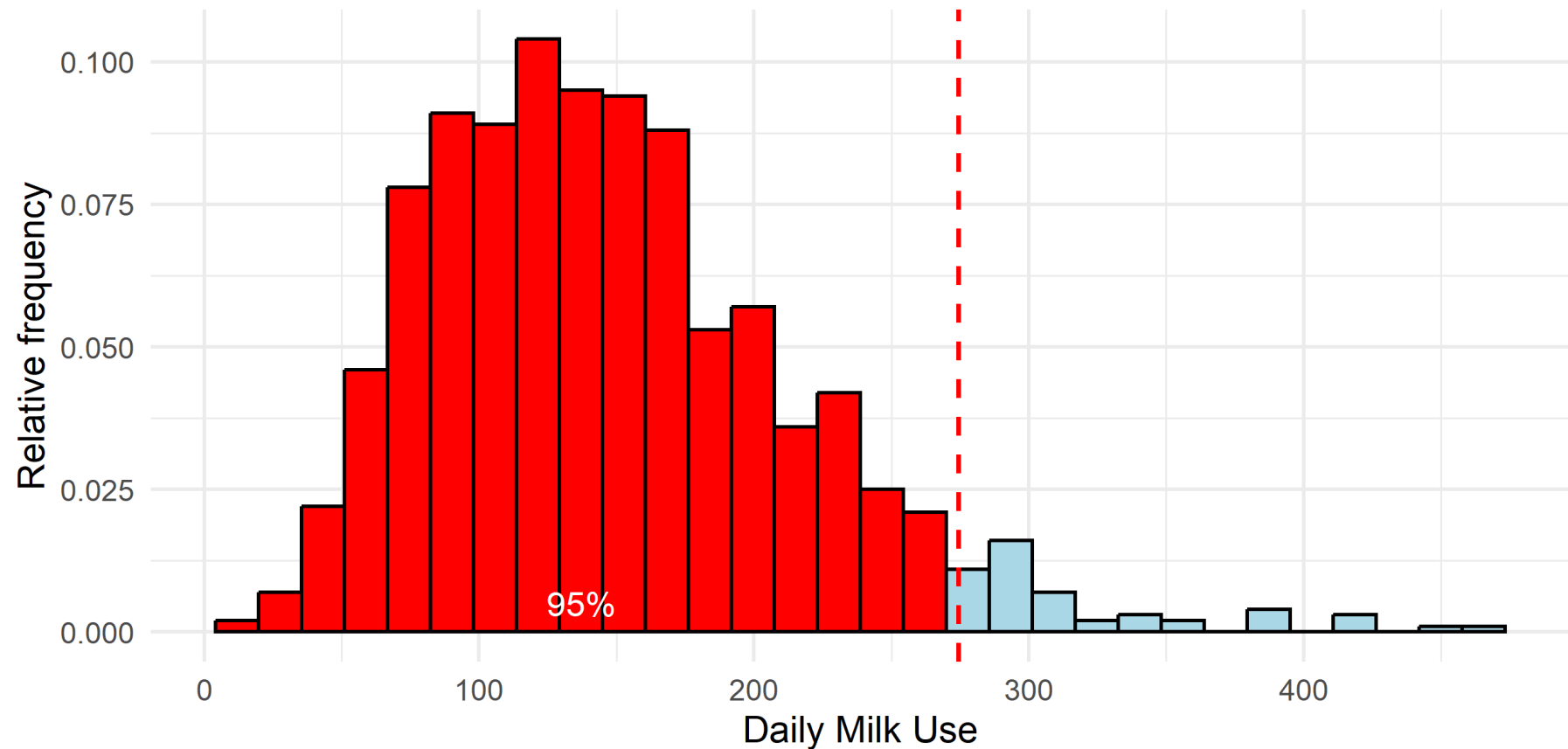
Percentiles (cont.)

Distribution of the daily use of milk:



Percentiles (cont.)

- Let s_i be the daily sales of milk
- We want to choose amount M , such that $P(s_i \leq M) = 0.95$
- That is, in 95% of days sales are smaller or equal than M



The answer is the **95th percentile** of the distribution (274 liters).

Percentiles — definition

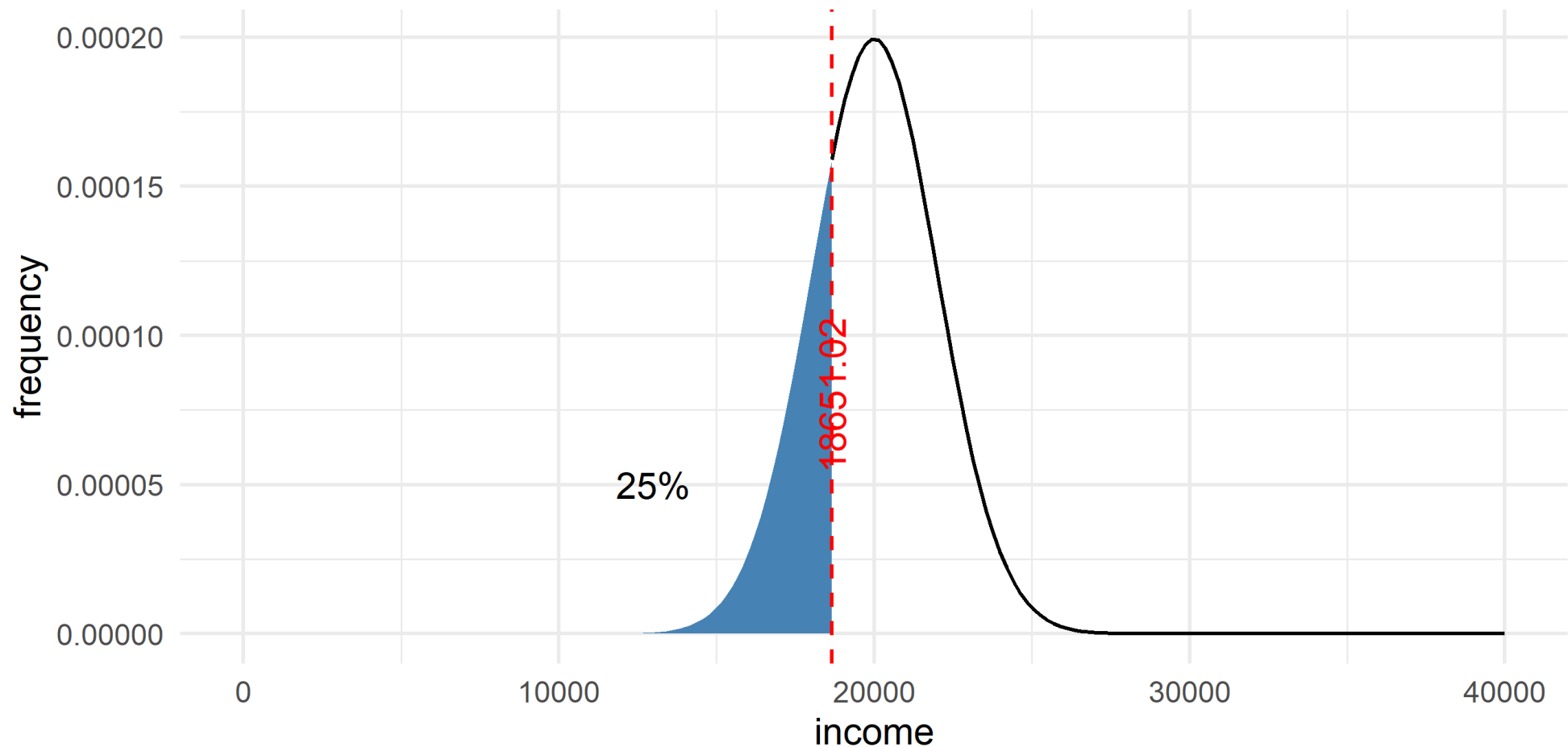
- *Percentiles* divide the ordered data into 100 equal parts.
- p th percentile is a value such that $p\%$ of the data are below it
 - v_p is such that $P(x_i \leq v_p) = p$
 - v_{95} is such that $P(x_i \leq v_{95}) = 95\%$
- What is the height such that 75% of ITAM students are smaller than this height?
- What is the income level such that 25% of people in Mexico earn less than that level?
- When do I need to leave home to be on time for class 90% of the days?

How to find it in a sample

1. Arrange the data in ascending order
2. Find which observation corresponds to the relevant percentile
 - Formula: $i = \left(\frac{p}{100}\right) (n + 1)$
 - Example: 95th percentile in a sample of 1000 observations $\rightarrow i = \left(\frac{95}{100}\right) (1000 + 1) = 950.95$
3. If it's an integer, the value of the i -th observation is your percentile
4. If it's not, take the average between i rounded down and i rounded up
 - Here: the average of the 950th and 951st observation

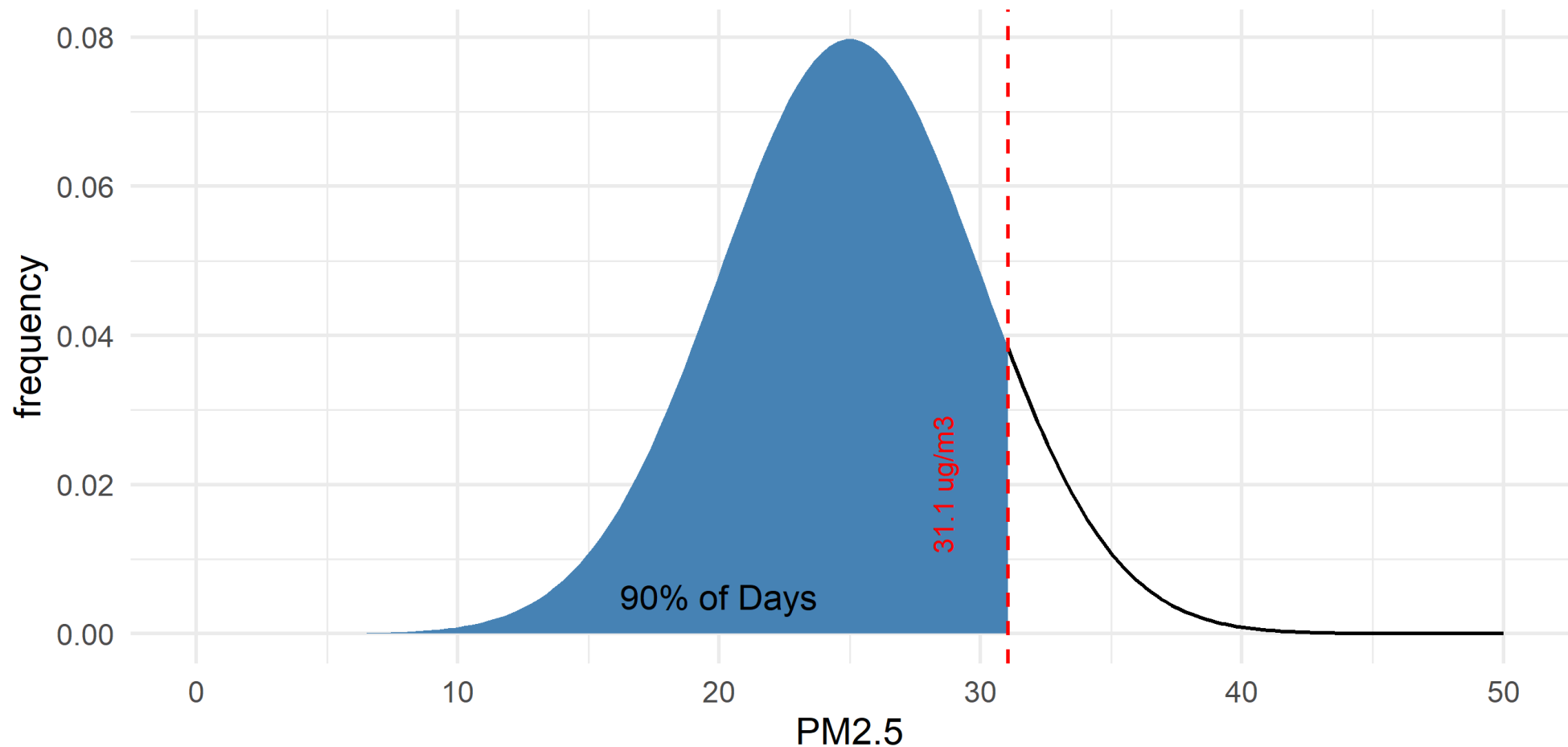
Common values

- **Median** — 50th percentile — half of the values are below the median
- **Quartiles** — 25th, 50th and 75th percentile.
 - How poor is the poorest quartile of the society? Their income is below the 25th percentile.



Common values (cont.)

- Deciles — 10th, 20th, ... 90th
 - How bad does pollution get in CDMX during the top 10% polluted days?
 - During the top 10% of polluted days, pollution is larger than the 9th decile.





Exercise 10a — Percentile by hand

Daily milk use (liters) in a café over 9 days. Find the **25th percentile** with the formula: sort x , compute the index $i = (25/100) * (\text{length}(x) + 1)$, then average the two sorted values around it.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 x <- c(120, 85, 190, 140, 95, 160, 210, 130, 105)
2 # Step 1: sort the data
3
4 # Step 2: compute the index i
5
6 # Step 3: i is not an integer -> average the 2nd and 3rd sorted values
```



Exercise 10b — Percentiles on real data

Compute the 10th, 25th, 50th, 75th, and 90th percentiles of `monthly_income` with `quantile()`. The 90th is the income 90% of workers earn below.

Exercise

↺ Start Over

⚠ Show Solution

▶ Run Code

1 # Your code here



Exercise 10c — Percentiles shape the distribution

Draw a histogram of `monthly_income` and add vertical lines at those five percentiles with `abline()`. Between any two neighboring lines lies a **fixed share of workers** — lines are packed where data is dense, spread out in the thin right tail.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 # Your code here
```

Exam spotlight — Fall 2025

Midterm 1

Real questions from last year's Midterm 1. **We solve these together in class.**

→ [More exam practice at the end of this deck](#)

Exam question — Fall 2025 Midterm 1

Households & Environment Survey, each household surveyed once in 2017.

Q1 (Dataset type). Each household appears once, in 2017. Is the data cross-sectional, time-series, or panel?

Q2 (Variable classification). Classify each: *Household size* · *Monthly electricity bill* · *Separates trash (yes/no)* · *Locality size (village < town < city)*.

Exam question — Fall 2025 M1 (cont.)

Q3 (Visualization). *An energy company wants to understand the distribution of weekly household gasoline spending to design a fuel-discount program.*

Visualize `gasoline_spending_weekly` and describe its shape.

Q5 (Boxplot — the median). A boxplot compares **monthly electricity bills** for households that use electricity for cooling vs those that do not. Which group has the **higher median** electricity bill?

The **median** is the line inside the box — compare the two lines.

Exercise 11 — The whole week, one question

You are the compensation analyst. Your manager asks:

“Do women in this sample earn less than men — and by how much?”

Answer it with the tools of this week. Decide what to compute, compute it, and build one picture that supports your answer.

Exercise

 Start Over

 Show Hint

 Run Code

```
1 # 1. A number: the typical income of each group. Which measure -- and why that one?
2
3 # 2. A picture: something a manager can read in three seconds.
4
5 # 3. One sentence: what do you tell your manager?
```

What you must be able to do this week

Data types & visualization

- Identify the **unit of observation** and whether data is **cross-sectional**, **time-series**, or **panel**.
- Classify a variable as **categorical (nominal/ordinal)** or **numerical (discrete/continuous)**.
- Build and read the right **chart** for a categorical variable (bar over pie).

Central tendency

- **Choose and compute** the **mean**, **median**, and **mode**, and know **which measure an outlier moves**.
- Compute the mean of a **binary** variable (a proportion).
- Find and interpret **percentiles**, **quartiles**, and **deciles**.

Next week: from *where the center is* to *how spread out the data is*.

Exam practice

These are the Week 2 exercise-set problems, on **real data**: an excerpt of the household survey used in last year's Midterm 1, real daily transit ridership, and ten years of Mexican labour-survey averages. Run the commands, look at what comes back, then decide.

[← Back to the exam spotlight](#)



Exam practice 1 — Data structures

Three **real** datasets are loaded: `households`, `ridership`, `wages_panel`. Each is a different structure. Classify them — **cross-sectional**, **time-series**, or **panel**. [D1]

Don't guess from the names. The answer is always the same two counts: **how many units** and **how many time points per unit**.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 # Look at each one first:
2 str(households)
3
4 # Then count units and time points. For example:
5 #   how many distinct years does wages_panel cover?
6 #   how many regions per year?
```



Exam practice 2 — Variable types

`households` is an excerpt of the **actual Midterm 1 dataset**. Classify its variables as **categorical**, **discrete**, or **continuous** — and get the evidence out of R before you answer. [D2]

Start with `household_size`, `electricity_bill_monthly`, `separates_trash`, `locality_size` — the four the exam asked about.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 str(households)
```

```
2
```

```
3 # Then dig into single columns:
```

```
4 #   table(households$<col>)           few repeated labels?
```

```
5 #   length(unique(households$<col>))  few whole numbers, or hundreds of values?
```



Exam practice 3 — Variable types II

Same three-way choice, from the description alone. [D3]

(a) Daily number of Uber trips completed by one driver in Mexico City.

→ **Discrete** — a count of trips.

(b) Whether an employee received job training during the past year.

→ **Categorical** — a yes/no label.

(c) Time (in minutes) to serve a customer in a bank branch.

→ **Continuous** — a measurement; any positive value.

(d) Number of times an individual has changed jobs in their career.

→ **Discrete** — a count.

Exam practice 4 — Does the sampling strategy lie?

Monterrey wants the share of restaurants with health-inspection violations. Inspecting all of them is too costly, so they must sample. For each strategy, is there **sample selection bias**? [D4]

(a) Every restaurant in one neighborhood. (b) A random 20% of the city. (c) All restaurants whose street number ends in 7. (d) The 40 largest by seating.

→ (a) **bias** — one neighborhood is not the city. (b) **no bias** — random sampling, the best strategy. (c) **no bias** — the last digit is unrelated to violations, so it is effectively random. (d) **bias** — the largest restaurants differ from the rest.



Exam practice 4 (cont.) — See it on real data

Same four strategies, but now you can **check them against the truth**: `households` is your whole “city”, so you know the real answer.

Estimate the mean monthly electricity bill under each strategy and compare:

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 truth <- mean(households$electricity_bill_monthly)
2 truth
3
4 # (a) only the large cities
5 # (b) a random 20%
6 # (c) household_id ending in 7
7 # (d) the 40 biggest households
```



Exam practice 5 — Selection bias in the wild

Is there a selection-bias concern, and why? [D5]

(a) A pharma company recruits trial participants through newspaper ads and online forums.

→ **Bias.** People who answer ads differ systematically from the target population.

(b) A university surveys only students who visit the career center.

→ **Bias.** Career-center visitors are more job-focused than the average student.

(c) A supermarket chain samples receipts at random across all its stores.

→ **No bias** for its own customers — but it says nothing about people who never shop there.

(d) A restaurant judges its food by its online reviews.

→ **Bias.** Reviews are self-selected: very happy and very angry customers write them.

(e) A consulting firm surveys firms about profits; a third refuse to answer.

→ **Bias** (non-response). Firms that answer are likely the more successful ones.